

OSCAR RAMÍREZ DE ARELLANO CASTELLANOS

Full Stack & DevOps Engineer · AI Integration & Vertical SaaS Architect

Veracruz, México · Open to remote roles worldwide · Native Spanish · Professional English

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PROFESSIONAL SUMMARY

Full-stack engineer with **20+ years shipping production software**, the last three focused on **AI integration and vertical SaaS for regulated industries** — fiscal/tax, legal, healthcare, and government. I take products from zero to working MVP in days, then scale them: multi-tenant auth, RAG pipelines, document AI, EMR/FHIR integrations, and the Mexican fiscal stack (SAT, CFDI 4.0, FIEL). Comfortable as a solo contributor, tech lead, or fractional CTO. Builder mindset, pragmatic engineering, monetization-first.

CORE EXPERTISE

AI & LLM Integration — OpenAI (GPT-4, Whisper, Vision), Anthropic Claude (Sonnet/Opus), Google Gemini, RAG pipelines (Qdrant, ChromaDB, Pinecone, Supabase Vector), Pydantic AI, LangChain, agentic systems, prompt engineering, multimodal OCR

Languages — TypeScript, JavaScript, Python, PHP, C#, SQL, Bash

Frontend — React 18/19, Next.js 14/15/16, Vue 3, Vite, Tailwind CSS, Material-UI, IBM Carbon Design, shadcn/ui, Chakra UI, Bootstrap; SPA & PWA development

Backend — Node.js (Express, NestJS), FastAPI, Django; REST API design, async processing, microservices

Databases & ORMs — PostgreSQL, MySQL, MS SQL Server, MongoDB, Redis, SQLite, Supabase, Firebase; Prisma, SQLAlchemy, Sequelize

DevOps & Cloud — AWS (EC2, S3, Route 53, Amplify), Vercel, Netlify, Docker, Nginx, PM2, Jenkins, GitHub Actions, cPanel/WHM, HashiCorp Vault, Tailscale, Linux server administration

Auth & Identity — Clerk, Auth0, Authentik, NextAuth v5, Azure AD, JWT, OAuth, RBAC

Healthcare Integrations — HL7 FHIR R4, HL7v2, HAPI FHIR, Mirth Connect, OpenEMR

Mexican Fiscal & Legal Tech — SAT Web Services (mass download), CFDI 4.0, FIEL/e.firma handling, AES-256-GCM credential storage, ISR/IVA/IEPS calculations, LFT compliance

Integrations & Automation — n8n, Zapier, Twilio, WhatsApp Business, Stripe, Salesforce, Mailchimp, SendGrid, Resend, Payload CMS

Testing & Observability — Vitest, Jest, Playwright, PostHog, Logfire

SELECTED PROJECTS

Vertical SaaS for Mexican Accounting Firms (2025–Present) Multi-tenant platform with full SAT mass-download state machine, CFDI 4.0 parsing (IVA/ISR/IEPS retentions, complemento de pago), encrypted FIEL credential storage, ISR/IVA calculations, and Claude-assisted fiscal analysis. Next.js 16, Prisma 6, PostgreSQL, NextAuth v5, Anthropic Claude API, @nodecdfi SDK, node-forge, ExcelJS.

Agentic RAG for US Consumer-Rights Law Firm Built three specialized Pydantic AI agents — refund calculator (Song-Beverly Act), compliance checker, and NHTSA/NMVTIS title verifier — integrated with Filevine case management, Slack, and a private web app. Self-hosted Qdrant with hybrid search, Claude (Sonnet for retrieval, Opus for synthesis), attorney-client-privilege-grade RBAC and audit trails. n8n orchestration, Docker, Nginx.

EMR Platform for US Clinical-Genetics Company Forked and customized OpenEMR with modules for fax/SMS, prior authorization, e-prescribing (Weno), and 8x8 voice integration. Delivered a modern Next.js 15 frontend with IBM Carbon Design System, patient portal with Clerk auth, FHIR R4 API, and security-audit-ready architecture (security assessment + implementation docs).

Wearable Hemodynamic Device → EMR Integration Designed and deployed a fully Dockerized integration stack: HAPI FHIR R4 server, Mirth Connect for HL7v2, n8n orchestration, Vault for secrets, Tailscale networking, AWS EC2. Real-time nurse-station dashboard with D3 charts and Next.js frontend.

KYC Compliance Platform for Mexican Customs Brokers Full-stack platform with risk-scoring engine (0–100, YAML-defined rule packs), AI extraction across 25+ fiscal/legal/customs document types, news scraper with AI analysis, and RBAC (Admin/Analyst/Client). FastAPI + SQLAlchemy 2.0, OpenAI GPT-4, Vite/React frontend, Docker.

Government Beneficiary Capture System (Mexico) PWA for social-program field operations with offline-first IndexedDB, real-time desktop↔mobile sync via Socket.io, hybrid OCR (OpenAI Vision + Tesseract.js fallback), CONEVAL poverty-index scoring, and RENAPO integration. Express + React + PostgreSQL + Zustand.

Modular Transcription SaaS Suite (Tranzcribe) Seven-component product: marketing site, B2C app, B2B app, admin dashboard (backend + frontend), and a white-label template stack (backend + frontend) for enterprise clients. OpenAI transcription, FFmpeg audio pipeline, Stripe billing, Clerk auth, Redis sessions, multi-provider email (Mailchimp / SendGrid / Resend).

Document Processor SaaS (DocuMail AI) Email-to-data pipeline: customers send a PDF, receive structured JSON/Excel back. Async PDF and Office parsing with pdfplumber, pytesseract, python-docx; LLM extraction (OpenAI + Anthropic); Stripe billing; aiosmtplib for email ingestion; React 19 + Vite frontend.

Full portfolio (80+ repositories) at github.com/OscarRamirezdeArellano

PROFESSIONAL EXPERIENCE

Senior Full-Stack Engineer & AI Integration Lead

IQ Soluciones Integrales en Tecnología (IQsit) · 2024 – Present · Remote

- Architected and shipped multiple AI-powered SaaS products for B2B and B2C — document processors, transcription platforms, fiscal/tax tooling, AI website generator, and HIPAA compliance assistants.
- Designed and deployed RAG pipelines on FastAPI with Qdrant and ChromaDB; multimodal extraction combining OCR (Tesseract, OpenAI Vision) with LLM structured-output prompting.
- Built multi-tenant SaaS architectures with Clerk, Auth0, NextAuth v5, and custom RBAC; subdomain-based tenant isolation, encrypted credential storage (AES-256-GCM), audit trails.
- Owned full DevOps lifecycle: AWS (EC2, S3, Route 53), Vercel, Docker, Nginx, PM2, Jenkins, cPanel/WHM, Vault, Tailscale; from VPS provisioning to production deploys.
- Orchestrated business workflows with n8n integrating OpenAI, Anthropic, Stripe, Twilio, WhatsApp Business, Mailchimp, and Salesforce.
- Acted as solo full-stack contributor and as fractional tech lead across simultaneous client engagements.

Senior Full-Stack & DevOps Engineer

Developers.NET · 2022 – 2024

- Delivered web applications on modern JavaScript stacks (React, Node.js, Next.js) and Python backends (Django, FastAPI) for clients across LATAM and USA.
- Implemented CI/CD pipelines (Jenkins, GitHub Actions, NGINX) and AWS-based hosting for production workloads.
- Introduced AI-powered automations and reporting tools when LLM APIs were still emerging — early adopter of OpenAI GPT integration in production systems.

Lead Developer & Systems Administrator

Rodall Oseguera, S.C. · 2010 – 2022

- Led development and project management for internal and client-facing applications in the international-trade and customs sector.
- Designed secure, robust platforms handling sensitive operational and financial data; managed full Linux server infrastructure, mail servers, and domain administration.
- Stack: Delphi, C#, PHP, Perl, JavaScript, MySQL, MS SQL Server, Firebird.

Earlier Career (2000 – 2010)

Software developer and webmaster roles across multiple companies — **SmartSolutia** (Senior Developer, custom ERP modules in PHP/C#.NET), **Tiburón Software** (Java/PHP business apps, Crystal Reports), **Universidad Empresarial** (Professor of Computer Vector Design; built academic-grading platform in Perl), **Rodall Oseguera** (initial tenure: web + multimedia in Clipper, Delphi, PHP, Flash), **Arrecife Dive Service** (Webmaster). Foundation in classical web stacks and early systems programming.

EDUCATION

Bachelor's Degree, Administrative Computer Systems · Universidad Veracruzana · 2004 – 2008

Bachelor's Degree, Electronic Engineering · Instituto Tecnológico de Veracruz · 2000 – 2004

Computer Technician (Vocational) · CBTIS No. 79 · 1997 – 2000

HONORS & RECOGNITION

- **1st Place National** — Software Development, *III National Competition of Specialties* (VI Academic Festival, DGETI)
- **Veracruz State Representative** — National CONACIBA-HUM-ES 2000, Software Development
- **State-Level Recognition** — Software Development, Xalapa, Veracruz (1996)

LANGUAGES

Spanish — Native · **English** — Professional working proficiency

References, demos, and additional project details available on request.